



Modular Athletic Exoskeleton Project Sensor Enclosure Placement Explorations- Report II

The document below shows new PCB placement based on research into biomechanical motion capture systems, testing the ergonomics of PCB housings, and adjusting placement based on those multiple points.

INTRODUCTION

Further research indicated that for optimal performance, each rigid body of importance needs **at least 2** sensors along the *length* of the limb, and that all joints of interest should **ideally** have **at least** one sensor. However, the position of joints can be estimated as the meeting point between two rigid bodies. For example, if the upper arm and lower arm rigid bodies are simplified as two lines and then extended until they meet, the elbow can be estimated as the meeting point. This warrants reconsidering PCB placement. Additionally, PCB housing prototypes are now available, adding more things to consider and test, as well as documenting the housings themselves.

In the last report, two strap systems were tested, and a third pocket system was not. The third system will be considered and tested within current capabilities, and the first two will be retested with the new PCB placement.

PCB HOUSINGS

SLS 3D print of the circuit boards was made to test the housing (Figure 1). They did not fit because of the rounded corners in the housing. The corners were cut to allow testing to continue. Scaling the housing slightly larger is recommended. The flat surfaces sit comfortably against the skin, with the slight exception of the shoulder.

PBC Placement Research and Results

Placing a PCB directly on the knee or elbow is a difficult task, but could theoretically be achieved. For the purposes of prototyping, it was not attempted. However, additional PCBs were placed on the shoulder and hip. Some placements had to be adjusted (discussed later in the document). A total of 20 node PCBs and one Main PCB are used. The main PCB is still in the middle of the chest, on the sternum, not far above the solar plexus. There are 2 nodes on the lower leg, 2 nodes on the lower leg, 1 on the hip, 1 on the shoulder, 2 on the upper arm, and 2 on the lower arm, making 20 in total (10 per side of the body).

PCB Placement and Testing

To test the placement of the PCBs, holes were cut into the bodysuit at various locations, and the lids of the PCB housings were extended (Figures 1, 2). The PBC then sticks out of the suit, while the fabric holds the housing to the body via the extended lid (seen in later Figures).

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Figure 1

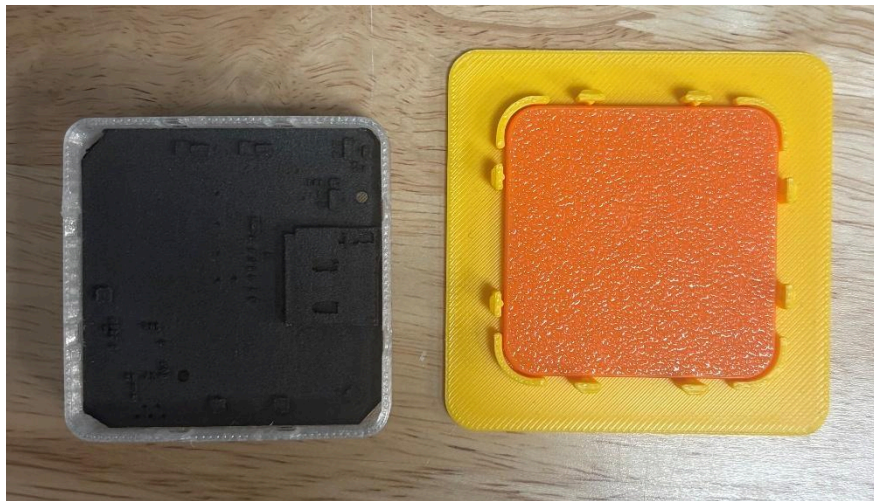


Figure 2

Beginning with the upper torso, the main PCB stayed where it was initially placed. The shoulder joint PCBs were initially placed on the front of the shoulders (Figure 3) to avoid issues when the arm was extended over the head. However, when the arm was brought across the front of the body (pectoral flexion), the shoulder PCBs got in the way (Figure 4).

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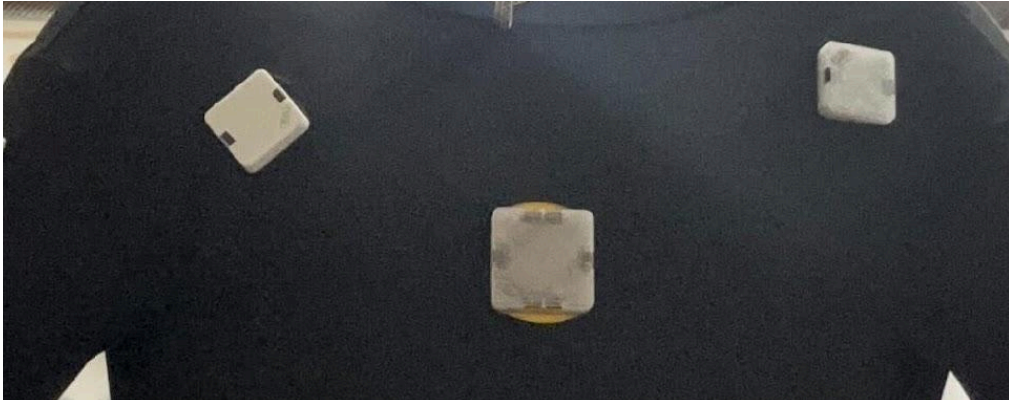


Figure 3



Figure 4

To resolve this issue, the PCB was placed between the top and back of the shoulder, ensuring to accommodate for full ROM (Figure 5, 6)

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Figure 5

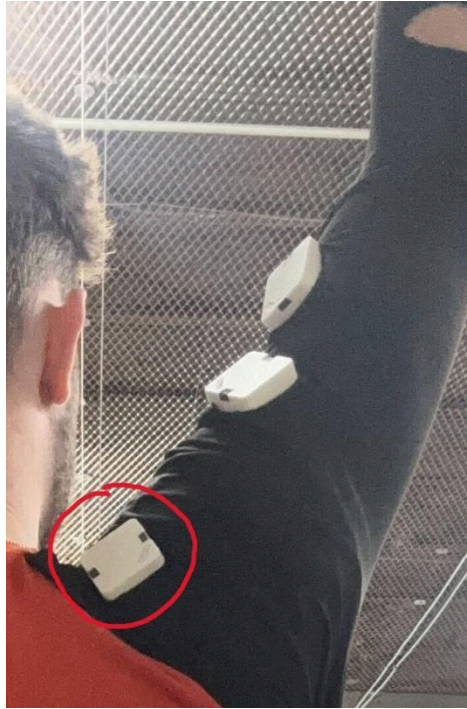


Figure 6

Moving to the arms, one additional PCB was needed to ensure the correct modeling of the upper arm when the product is in use. It is also on the outside of the bicep, closer to the shoulder than the first (Figure 7, 8); full ROM was still tested.



Figure 7



Figure 8

Moving down, additional PCBs were placed on the sides of the hips (Figure 9). This helps with the accuracy of leg modeling as well as properly framing the torso.



Figure 9

Next, the leg was tested. An additional PCB was placed on the upper leg, similarly to the upper arm. Initially, the PCBs for the upper leg were placed on the front/top of the thigh (Figure 10), but after review of existing technologies, the side of the leg was deemed more optimal (Figure 11,12). It both keeps the PCBs out of the way in the case of extreme frontal hip flexion (bringing the knees to the chest) and puts them more consistently in line with the lower leg PCBs (Figure 12). Once again, the full ROM was tested.



Figure 10



Figure 11



Figure 12

Final/Additional Notes

A full suit with holes only cut in the desired locations was made and tested in the same way as the one in this report, with promising results. For convenience, either a strap to allow the user to successfully use the zipper with less difficulty (commonly seen in wetsuits) or a new closure system is recommended. Padding in the groin and chest area is also recommended for comfort and modesty purposes.

The full suit, despite seeming to be a breathable material, got warm quickly. Another similar product of a higher quality, perhaps specifically marketed towards athletes (similar to what is commonly used by dancers and stage actors), should be investigated as a possibility further in the design process.

This attachment method for PCBs, while initially only intended for prototype purposes, could (with some adjustment) be suitable for production. Strengthening where the holes were cut and sewing a new barrier between the PCB and the skin, having the housing latch through the clothing without making cuts, or another method inspired by the one used in testing could be useful in saving on material cost and seamster services.

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